

$$1. a) \lim_{n \rightarrow \infty} \left(\frac{2^{n+1}}{2^n} \right) = \lim_{n \rightarrow \infty} \left(\frac{2 \cdot 2^n}{2^n} \right) = 2 \leftarrow \text{constant}$$

$$\boxed{2^{n+1} = O(2^n)}$$

$$b) \lim_{n \rightarrow \infty} \left(\frac{2^{2n}}{2^n} \right) = 1 \leftarrow \text{constant}$$

$$\boxed{2^{2n} = O(2^n)}$$

$$2. a) \lim_{n \rightarrow \infty} \left(\frac{(4/9)^n}{(4/9)^n} \right) = 1 \leftarrow \text{constant}$$

$$\boxed{\Theta = (4/9)^n}$$

$$b) \begin{array}{ccc} n \lg(n) & -15n & +14\sqrt{n} \\ \downarrow & \downarrow & \downarrow \end{array}$$

$$n \lg(n) + 15(n \lg(n)) + 14(n \lg(n)) = 30n \lg(n)$$

↑ remove constant

$$\boxed{\Theta = (n \lg(n))}$$